

Project Report: Ferns and Petals Sales Analysis Dashboard

Project Overview

This project focuses on analyzing sales data for Ferns and Petals, a company specializing in gifts and floral products. The objective is to provide valuable insights into revenue trends, customer behavior, and product performance using **Excel's Power Query, Power Pivot, Pivot Tables, and Data Visualization Techniques**.

Objective

The primary objectives of this project are:

- To understand the **total revenue and customer spending patterns**.
- To identify **top-performing products and categories**.
- To analyze **sales trends by occasions, months, and order timing**.
- To evaluate **geographical sales distribution**.
- To provide **data-driven insights for business decision-making**.

Data Sources and Methodology

Data Sources

The project utilizes data from three primary tables:

- **Products Table:** Contains details of various gift items.
- **Orders Table:** Includes information on orders, revenue, and customer purchases.
- **Customers Table:** Provides demographic insights into customer behavior.

Tools and Techniques Used

- **Power Query & Power Editor:** Data cleaning, transformation, and structuring.
- **Power Pivot & Measures:** Data aggregation and complex calculations.
- **Pivot Tables & Charts:** Dynamic data visualization and analysis.
- **Slicers & Filters:** Interactive selection for better insights.

Key Insights & Findings

Revenue Analysis

- **Total Revenue:** ₹35,20,984.00
- **Average Customer Spending:** ₹3,520.98

- **Total Orders Processed:** 1,000
- **Average Delivery Time:** 6 Days

Sales Trends

- **Revenue by Occasions:** Highest sales were observed during **Raksha Bandhan, Valentine's Day, and Anniversaries.**
- **Revenue by Month:** Peak sales were recorded in **March and June**, while a decline was observed in other months.
- **Revenue by Hour:** Sales fluctuated throughout the day, with peaks during specific hours.

Product Performance

- **Top Revenue-Generating Categories:** Colors (₹10,05,645), Soft Toys, and Plants.
- **Top-Selling Products:** Magman Set, Harum Pack, and Quia Gift.

Geographical Insights

- **Top Cities by Orders:** Dhanbad, Guwahati, and Indore had the highest number of orders.
- **Sales Distribution:** Orders were widely distributed across multiple urban locations

Business Impact & Recommendations

Business Impact

The dashboard provides a clear overview of sales performance, helping the business in:

- **Optimizing marketing campaigns** based on seasonal trends.
- **Enhancing inventory management** for top-selling products.
- **Improving operational efficiency** by reducing delivery times.
- **Targeting high-revenue cities** for better customer engagement.

Recommendations

- **Marketing Focus:** Increase promotional efforts around **high-sales occasions** such as Raksha Bandhan and Valentine's Day.
- **Inventory Planning:** Ensure **sufficient stock levels** for high-demand product categories.
- **Customer Retention:** Offer personalized discounts based on **customer purchase behavior.**

Conclusion

This sales analysis dashboard provides a **data-driven approach** to understanding customer behaviour and optimizing business strategies. With clear revenue trends, product performance insights, and geographical sales distribution, the dashboard acts as a powerful tool for **strategic decision-making** and business growth.

Next Steps: Future improvements could include **advanced predictive analytics**, integration with **real-time sales data**, and the incorporation of **machine learning techniques** for deeper customer insights.

Visuals & Charts Used

- Bar Charts for **Revenue by Occasion and Product Category**.
- Line Graphs for **Revenue Trends by Month and Hour**.
- Column Charts for **Top-Selling Products and Cities by Order**.